# $\mu \to e\gamma$ in the $S_3$ 3HDM: forbidden at the exact vacuum, and the first LFV constraint that bites on the scalar sector

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**Abstract**

I have computed $\mathcal{B}(\mu \to e\gamma)$ at all 2021 viable points of the soft-broken scan, for every allowed value of the lepton parameter $\mu_3^\ell$ and every branch of the charged-lepton fit. That is 415 464 configurations, compared with the MEG II limit $1.5 \times 10^{-13}$ [1]. The amplitude sums the one-loop diagrams of all seven physical scalars and the Barr–Zee two-loop diagrams with top, bottom, $\tau$, $W$ and charged-Higgs loops. Each piece is checked against an independent source. Three results stand out. First, at the exact-$S_3$ vacuum the rate is *identically zero*: a residual $\mathbb{Z}_2$ forbids it to all orders, so $\mu \to e\gamma$ measures the soft breaking directly. Second, in the soft vacuum it is by far the strongest LFV constraint. Only **610 of 2021** scalar points keep an allowed lepton configuration, where the Higgs LFV decays keep 2016. Third, the $W$-loop Barr–Zee diagram through the SM-like Higgs dominates, so the survivors are the points closest to the alignment limit. The conclusion survives the two weakest parts of the amplitude, the quark couplings and a 14% normalization difference in the $W$ loop.

## 1 Summary

Everything below comes from `research/thdm_s3/07_mu_to_e_gamma.ipynb`, with the numerics in `radiative.py`, `soft_decays.py` and `lfv.py`.

1. **The couplings** of every scalar come off the Lagrangian and agree with `feynlag`'s extractor (§2).
2. **The amplitude** is assembled from three papers and checked piece by piece (§3). One of them has a typo, and the charged-Higgs Barr–Zee term is built here.
3. **Exact $S_3$: zero** (§4).
4. **Soft vacuum: it removes about 70% of the scalar points** (§5).

The previous report found $h \to e\mu$ to be the binding LFV Higgs decay, excluding only 5 points. The reason $\mu \to e\gamma$ is so much stronger is standard [3]: a Barr–Zee diagram needs only one flavour-violating coupling, combined with the large $hWW$ and $htt$ couplings, and MEG II is more sensitive than any Higgs measurement.

## 2 Couplings of every scalar

$h \to \ell\ell'$ needed only the CP-even couplings. Here all three kinds enter. Expanding each doublet as $H_k = \big(c_k, (v_k + r_k + i a_k)/\sqrt{2}\big)$ in the five-term Yukawa Lagrangian and differentiating gives per-doublet matrices in the convention $L \supset -\bar{e}_L Y^{\phi} e_R\, \phi - \bar{\nu}_L Y_C e_R\, H^+ + \text{h.c.}$ of [2, 3]. The doublet structure fixes

$$Y^{(a_k)} = i\, Y^{(r_k)}, \qquad Y^{(c_k)} = \sqrt{2}\, Y^{(r_k)},$$

and the computation reproduces both relations. The physical couplings follow from the rotation columns. Unlike $h \to \ell\ell'$, the dipole needs the *signs* of the lepton eigenvalues, because its $\tau$-enhanced term $m_\tau Y^*_{\tau e} Y^*_{\mu\tau}$ flips sign with the $\tau$ eigenvalue. The right-handed fields are rephased to make every mass positive.

| check | against | agreement |
|---|---|---|
| all 7 scalars' $3 \times 3$ couplings, soft point | `feynlag` extractor | $10^{-18}$ |
| Goldstone couplings flavour-diagonal | $i\,\mathrm{diag}(m)/v$, $\sqrt{2}\,\mathrm{diag}(m)/v$ | $10^{-15}$ (102 points) |

The Goldstones carry no flavour violation at any vacuum. This is what allows them to be dropped (their part of the $W$ loop is already inside the Barr–Zee $W$ term).

## 3 The amplitude

With $T = e\,\varepsilon^*_\alpha \bar{u}_e\, m_\mu i\sigma^{\alpha\beta} q_\beta (A_L P_L + A_R P_R) u_\mu$,

$$\mathcal{B}(\mu \to e\gamma) = \frac{48\pi^3 \alpha\, (|A_L|^2 + |A_R|^2)}{G_F^2} \qquad [2], \text{ Eqs. (28)–(29).}$$

[3]'s Wilson coefficients are $c = 4\pi^2 A$, and the two papers' one-loop results agree under this map, sign included. They then print the same Barr–Zee formulae, so one-loop and two-loop pieces can be added with a common sign.

### 3.1 One loop

[3] Eq. (A3) gives the neutral-scalar diagram as a two-parameter Feynman integral. With the external masses dropped from the denominator ($m_\mu^2/m_\phi^2 \lesssim 10^{-6}$) it integrates in closed form:

$$c_L = \sum_{f=e,\mu,\tau} \frac{\frac{1}{2} I_1(t_f)\,(m_e Y_{ef} Y^*_{\mu f} + m_\mu Y^*_{fe} Y_{f\mu}) + I_2(t_f)\, m_f Y^*_{fe} Y^*_{\mu f}}{4 m_\mu m_\phi^2}, \qquad t_f = \frac{m_f^2}{m_\phi^2},$$

$$I_1 = \frac{t^3/6 - t^2 + t\ln t + t/2 + 1/3}{(1-t)^4}, \qquad I_2 = \frac{t^2 - 4t + 2\ln t + 3}{2(t-1)^3},$$

and $c_R$ is obtained by $Y \to Y^\dagger$. This is exact in $m_f/m_\phi$, so a light scalar needs no expansion. It agrees with the full integral, external masses kept, to $10^{-8}$ at $m_\phi = 700$ GeV and $6 \times 10^{-6}$ at 25 GeV. For $t \to 0$ it gives the familiar $\ln(m_\phi^2/m_f^2) - \frac{3}{2}$.

**A typo in [2].** Their Eq. (30) multiplies the non-flip term $y_{i\mu}/6$ by $m_i/m_\mu$. That term takes its chirality flip from the external muon, so it carries $m_\mu$, not $m_\tau$. [3] Eq. (A3) and [2]'s own $\tau \to \mu\gamma$ formula agree with this.

The charged-scalar loop, with massless neutrinos, is $A_L = -(Y_C^\dagger Y_C)_{e\mu}/(192\pi^2 m^2_{H^\pm})$ and $A_R = 0$ [2]. This is $-\frac{1}{2}$ of the neutral non-flip term, and [5]'s general formulae give the same ratio independently.

### 3.2 Barr–Zee

The fermion and $W$ loops in the $\phi\gamma\gamma$ vertex are [2] Eq. (20), which is identical to [3] Eqs. (A4)–(A5).

| piece | reference value | this code |
|---|---|---|
| top, $\tau \to \mu\gamma$ via the SM Higgs | $-0.082\, Y_{tt}$ [3] Eq. (14) | $-0.0811$ (1%) |
| $W$, same | 0.11 [3] Eq. (14) | 0.1254 (14% above) |
| $W$ bracket, $m_\phi = 125$–$800$ GeV | $F^{(3)}$ of [4] | 1% |

The $W$ formula is the one both papers print, and it matches [4]'s independent $g-2$ calculation. The 14% gap is therefore with [3]'s *number*. The neglected $\phi Z\gamma$ piece moves it the wrong way (+6%), so the likely cause is [3]'s "set C" diagrams, which are not implemented. Once the gap is rescaled out, the full assembly reproduces [3]'s $\mu \to e\gamma$ bound on an SM-like Higgs to 2%. With MEG II that bound becomes

$$\sqrt{|Y_{\mu e}|^2 + |Y_{e\mu}|^2} < 7.2 \times 10^{-7}.$$

**The charged-Higgs Barr–Zee term** appears in neither [2] nor [3]. [4] gives both one-loop $\phi\gamma\gamma$ vertices with an off-shell photon. They have the same $k^2$-dependence, so the second loop is the same integral, and the $H^\pm$ term follows from the fermion term by the substitution $m_f y_{ff} N_c Q^2\,(1-2x(1-x)) \to (v\lambda/2)\,x(1-x)$. The sign of $\lambda$ relative to `feynlag`'s trilinear ($L \supset g\,\phi H^+ H^-$, $\lambda = -g/v$) is not assumed. It is fixed by the low-energy theorem, which requires the scalar-to-top ratio of the heavy-loop $\phi\gamma\gamma$ vertices to equal that of `feynlag`'s validated diphoton amplitude.

**Not included:** the $\phi Z\gamma$ Barr–Zee diagrams (suppressed by $1-4s_W^2$), the $H^\pm W^\mp\gamma$ ones, and [3]'s set C. The quark couplings are an **ansatz**. Each quark's Yukawa lies along the vacuum direction, so $y_{tt}(h_i) = \kappa_i m_t/v$ and the CP-odd states have none, as in the previous report's $R_{\gamma\gamma}$.

## 4 The exact-$S_3$ vacuum: identically zero

At 40 exact points of the first scan, evaluated at $\varphi = \pi/3$ with no soft terms, at three values of $\mu_3^\ell$, the rate is $\mathcal{B} \sim 10^{-30}$, which is round-off. This is not because the couplings are small: $h_0$, one $A$ and one $H^\pm$ carry $e\mu$ couplings up to $\sim 0.1$.

The reason is the **residual** $\mathbb{Z}_2$ of the aligned vacuum, the $S_3$ reflection that leaves it invariant. Under it the electron, which is the generation the $O_{12}$ rotation decouples, is odd, while $\mu$ and $\tau$ are even. The scalars orthogonal to the vacuum *within* the doublet plane ($h_0$, one $A$, one $H^\pm$) are odd, and the rest are even. A $\mu \to e\gamma$ amplitude has one odd external lepton and is forbidden while the $\mathbb{Z}_2$ holds. In particular $h_0$ has neither a $VV$ coupling nor a diagonal $h_0 H_a^+ H_a^-$ coupling, so its $e\mu$ coupling feeds no Barr–Zee diagram. This is the radiative-decay counterpart of the first-generation decoupling that forces $V_{us} = 0$ in the fermion-sector report.

So $\mu \to e\gamma$ **measures the soft breaking directly.** Moving one exact point off the alignment, with the free soft terms still zero, gives

$$\mathcal{B} \propto (60^\circ - \varphi)^2 \quad \text{(fitted slope 2.002)},$$

that is, an amplitude linear in the $\mathbb{Z}_2$ breaking. A misalignment of $0.1^\circ$ already gives $1.9 \times 10^{-13}$, at the MEG II limit.

## 5 The soft-broken scan

Every point × the 73-value $\mu_3$ grid of the previous report × every lepton branch gives 415 464 configurations. For each one the SM-like state's $h \to \ell\ell'$ rates are computed alongside, so the two constraints are compared configuration by configuration.

| | configurations allowed | scalar points allowed |
|---|---|---|
| three Higgs LFV limits [6, 7] | 79.8% | 2016 / 2021 |
| MEG II | 2.50% | **610** / **2021** |
| both | | 599 / 2021 |

The median rate is $\mathcal{B} = 7.7 \times 10^{-10}$. Configuration by configuration, measured against its own limit, $\mu \to e\gamma$ is a median $9.2 \times 10^4$ times more restrictive than $h \to e\mu$. Unlike every Higgs LFV decay, it **constrains the scalar sector**.

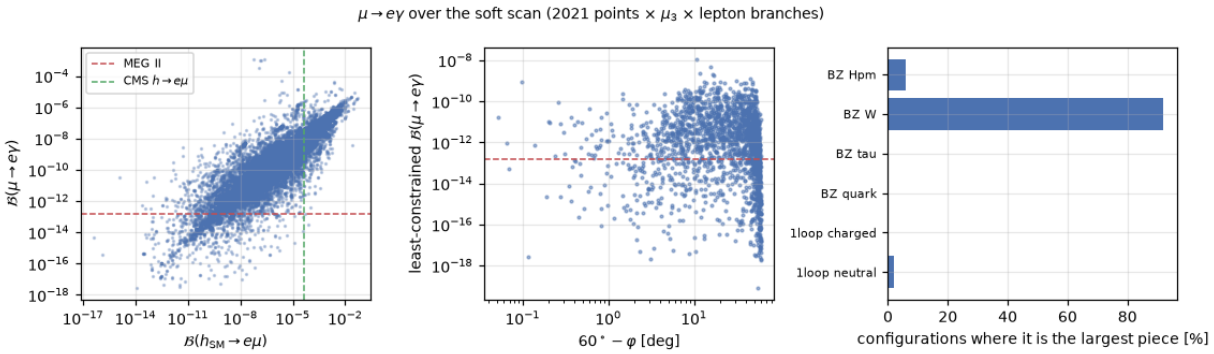



Figure 1: Left: $\mathcal{B}(\mu \to e\gamma)$ against the SM-like $\mathcal{B}(h \to e\mu)$ per configuration (40 000 sampled), with the MEG II and CMS limits dashed. Middle: the least-constrained configuration of each scalar point against the vacuum misalignment. Right: which piece of the amplitude is largest, by share of configurations.

**What drives it.** The Barr–Zee $W$ loop through the SM-like Higgs is the largest piece in 92% of configurations, the $H^\pm$ Barr–Zee loop in 6.0%, and the one loop in about 2%. The SM-like state's $e\mu$ coupling is pure non-vacuum admixture (previous report), so the survivors are the more aligned points: their median $1-\kappa_V^2$ is $8.3\times10^{-4}$, compared with $2.4\times10^{-3}$ overall. No $\mu_3$ window or lepton-branch kind is singled out. MEG-allowed configurations span the whole $|\mu_3|$ axis and all three branch kinds. Being close to $\varphi = \pi/3$ does not protect a point (Fig. 1, middle). In the scan the free soft terms break the $\mathbb{Z}_2$ whatever $\varphi$ is, so $\varphi$ is not its order parameter.

**Robustness.** The two least certain inputs are the quark couplings (an ansatz, and the top loop interferes destructively with the $W$ loop) and the 14% $W$-loop gap. Each configuration carries three variants built from the same amplitudes:

| variant | scalar points allowed by MEG II |
|---|---|
| baseline | 610 |
| no quark Barr–Zee | 499 |
| quark couplings with opposite sign | 456 |
| $W$ loop scaled to [3]'s 0.11 | 675 |

The count moves by about 25%, and the conclusion that MEG II removes most of the scalar sector does not.

## 6 Open questions

1. **Missing two-loop pieces:** [3]'s set C, and the $\phi Z\gamma$ and $H^{\pm}W^{\mp}\gamma$ Barr–Zee diagrams. The variants above suggest their effect on the point count is of the same order as the quark uncertainty, but they are not computed.

2. **The quark sector per point**, to replace the Barr–Zee quark ansatz. The same input is missing from $R_{\gamma\gamma}$.

3. **An order parameter for the residual** $\mathbb{Z}_2$ in the soft vacuum: a combination of soft terms rather than $\varphi$. It would turn the near-zero of §4 into a scan variable.

4. $\mu \to 3e$ **and** $\mu$–$e$ **conversion** share this dipole and add contact terms. $\tau \to \mu\gamma$ **and** $\tau \to e\gamma$ need only the flavour indices of `radiative.py` generalized.

## References


[1] MEG II Collaboration, *New limit on the* $\mu^+ \to e^+\gamma$ *decay with the MEG II experiment* (2025), arXiv:2504.15711.

[2] Y. Omura, E. Senaha and K. Tobe, $\tau$- *and* $\mu$*-physics in a general two Higgs doublet model with* $\mu - \tau$ *flavor violation*, Phys. Rev. D **94**, 055019 (2016), arXiv:1511.08880.

[3] R. Harnik, J. Kopp and J. Zupan, *Flavor violating Higgs decays*, JHEP **03** (2013) 026, arXiv:1209.1397.

[4] V. Ilisie, *New Barr-Zee contributions to* $(g-2)_\mu$ *in two-Higgs-doublet models*, JHEP **04** (2015) 077, arXiv:1502.04199.

[5] L. Lavoura, *General formulae for* $f_1 \to f_2\gamma$, Eur. Phys. J. C **29**, 191 (2003), hep-ph/0302221, doi:10.1140/epjc/s2003-01212-7.

[6] CMS Collaboration, *Search for lepton-flavor violating decays of the Higgs boson in the* $\mu\tau$ *and* $e\tau$ *final states in pp collisions at* $\sqrt{s} = 13$ *TeV*, Phys. Rev. D **104**, 032013 (2021), arXiv:2105.03007.

[7] CMS Collaboration, *Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the* $e\mu$ *final state in proton-proton collisions at* $\sqrt{s} = 13$ *TeV*, Phys. Rev. D **108**, 072004 (2023), arXiv:2305.18106.